

Triage sheet — machine attack on the strange-series AdS↔dS question of arXiv:2512.10101 (v2).

Candidate findings of an unreviewed machine-generated project (repository `oleg-n-pavlov/opmath`; statement numbers below refer to its `tex/main.tex`, 33 pp.). Asked of you: which items, if any, merit a closer look; ordered cheapest-to-check first. Tags: [PROVED]; [DERIVED] = calculated, not independently checked; [ARGUED]; [CONJECTURED]. Notation is your paper's, plus $\mu(y) = \frac{1}{2}(y + y^{-1})$.

1. **Two suspected misprints in v2 (4.19).** The Γ_{q^2} -argument shift is printed as $+i\pi/\ln q$, twice (exponent and middle member); consistency with (4.18) and the §4.3 substitution $\ell \mapsto \ell + \frac{i\pi}{2\ln q}$ requires $+i\pi/(2\ln q)$ — under the printed full-period shift the q -gamma ratio would not vanish. The vanishing conclusion is unaffected; (4.20) as printed is confirmed. [DERIVED] [`lit/notes/2512.10101.md` §7; note in Prop. 1.6]
2. **Exact finite- q reflection identity.** For all $q, \ell \in \frac{1}{2}\mathbb{Z}_{>0}$, θ_i : $\mathcal{K}_q^S(\theta_1, \theta_2) = \mathcal{K}_q^O(\theta_1, \pi - \theta_2)$ — the strange equal-time kernel is the discrete one with one argument reflected through the band centre. Hence (4.20) is an identity at every q , no limit needed; the (4.19) vanishing gets an explicit counting bound (per factor $(3/7)^N$, $N \sim \ln 2/(2|\ln q|)$, that part proved). [DERIVED] [Lemma 1.5, Prop. 1.6]
3. **Exact marginal line and band-centre crossing for complex-weight bilocals.** For chord weights t^k , $0 < |t| < 1$, the cross-edge/same-edge modulus ratio is $\equiv 1$ on $\arg t = \pm \frac{\pi}{2}$ at *every* q (and, as $q \rightarrow 1$: suppressed for $|\arg t| < \frac{\pi}{2}$, enhanced beyond, strange ray $\arg t = \pi$ extremal); on the principal circle $t = q^{1+2ib}$ the crossing sits at $b_{\max}/2$, spectral image $\theta = \frac{\pi}{2}$ exactly — the band centre, i.e. the centre-dS locus (2310.16994) and the sine-dilaton curvature flip (2404.03535). [PROVED] [Thm. 6.8, Lemma 6.7; Rmks. 7.5–7.6]
4. **q -Askey $\alpha < -1$ bound states sit at strange Casimir values.** The finite bound-state towers of the q -Askey-deformed chord Hamiltonians of 2605.13956 at deformation $\alpha < -1$ are exactly $\tilde{\Omega} = -\mu(|\alpha|e^{-2\lambda l}) < -1$, the strange range, log-periodically spaced (one line of arithmetic from their §2). Unremarked there; “apparently new” is this project’s own literature-search conclusion. [DERIVED] [Rmk. 7.4]
5. **Covariance obstruction.** Any assignment C constant on the $U_q(\mathfrak{su}(1,1))$ -constituents of each irreducible corepresentation of $SU_q(1,1) \rtimes \mathbb{Z}_2$ in the regular corepresentation satisfies $C(\pi_{\ell-1/2, \epsilon}^S) = C(D_\ell^+) = C(D_\ell^-)$ for all $\ell \in \frac{1}{2}\mathbb{Z}_{\geq 1}$, by GKK’s fusion $W_{p,x}|_{U_q} \cong \pi_{\ell-\frac{1}{2}, \epsilon}^S \oplus D_\ell^- \oplus D_\ell^+$: a criterion typing strange = dS against discrete = AdS cannot be normaliser-covariant. [PROVED] [Thm. 1.21; fusion transcribed from 0905.2830 as Thm. 1.16]
6. **Radial part of the Casimir on the reduced space.** Slice reduction leaves exactly two operators: the chord transfer matrix ($\chi\xi > 0$; no strange states there) and, on the other wedge, a Jacobi operator on $\ell^2(\mathbb{Z})$ with deficiency indices $(1,1)$ (an operator of Koelink, math/0305385; parameter identification ours). Each self-adjoint extension carries one discrete-type and one strange-type eigenvalue tower locked by $y_1 y_2 \in -q^{2\mathbb{Z}}$ (Abel’s theorem); the Π -invariant extensions occur exactly at the Plancherel labels $2a_0 \in \mathbb{Z}$, with spectrum $\{\pm\mu(q^{2a_0+2j})\}_j$ — GKK’s fusion pairing $a = \ell - \frac{1}{2}$ realised spectrally. [PROVED] (reduction, indices) / [DERIVED] (product rule, fusion extension) [Prop. 5.1, Thms. 5.7, 5.9, Cor. 5.10]
7. **Verdict on the open question.** Geometric form refuted: every intrinsic criterion is normaliser-covariant, hence obstructed by item 5 (realised concretely in item 6), and the one grading-breaking datum, the spectral orientation, places the *strange* tower at the AdS edge; no quantity crosses as a varies. Amplitude form holds under edge-dS (2411.16922 / 2505.08116) — it is items 2–3 — and fails under centre-dS (2310.16994). The AdS↔dS bridge the framework supports is the grading Π itself. [PROVED] (intrinsic class) / [DERIVED] (oriented and amplitude classes) [Thm. 6.5]

Status. No human has read any of this. The numerics (60-digit, committed outputs reproduce bit-for-bit) only self-check the project’s own formulas. Items 5 and 7 stand or fall with the covariance requirement stated in 5. The load-bearing inputs — the fusion theorem and the strange-series matrix elements — are transcribed from 0905.2830, not re-derived; and 0905.2830’s stated principal-series Casimir eigenvalue $\mu(q^{2ib})$ contradicts its own printed matrix elements ($-\mu(q^{2ib})$), resolved here numerically in favour of the matrix elements [Conv. 2.8]. All novelty markers are the project’s own literature search, nothing more.

The ask: which of 1–7, if any, is worth your time. “None” is a complete answer.